

UNIVERSITY OF MASSACHUSETTS
Dept. of Electrical & Computer Engineering

Introduction to Cryptography
ECE 597XX/697XX

Part 9

Elliptic Curve Cryptography

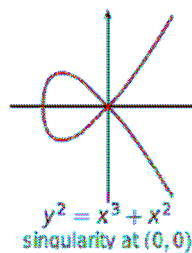
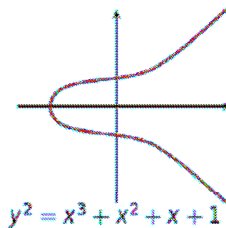
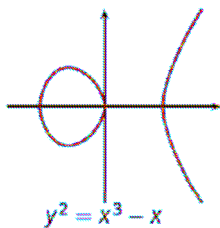
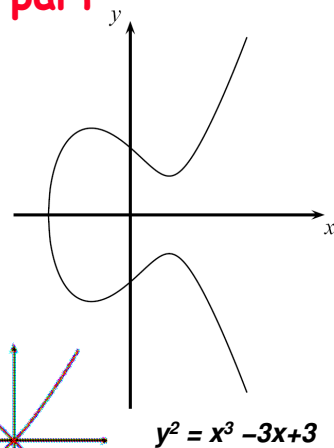
Israel Koren

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Adapted from Paar & Pelzl, "Understanding Cryptography," and other sources

Content of this part

- Introduction
- Computations on Elliptic Curves
- The Elliptic Curve Diffie-Hellman Protocol
- Security Aspects
- Implementation in Software and Hardware



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Adapted from Paar & Pelzl, "Understanding Cryptography," and other sources

Motivation

- **Problem:**

Asymmetric schemes like RSA and ElGamal require exponentiations in integer rings and fields with parameters of more than 1000 bits

- High computational effort on CPUs with 32-bit or 64-bit arithmetic
- Large parameter size is difficult (storage space) for small embedded devices

- **Goal:**

Smaller field sizes that can provide equivalent security

- **Solution:**

Elliptic Curve Cryptography uses a group of points (instead of integers) for cryptographic schemes with coefficient sizes of 160-256 bits, reducing significantly the computational effort

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Elliptic Curves over $\mathbb{R}$

- ◆ Elliptic curves are polynomials that define points based on the (simplified) Weierstrass equation:

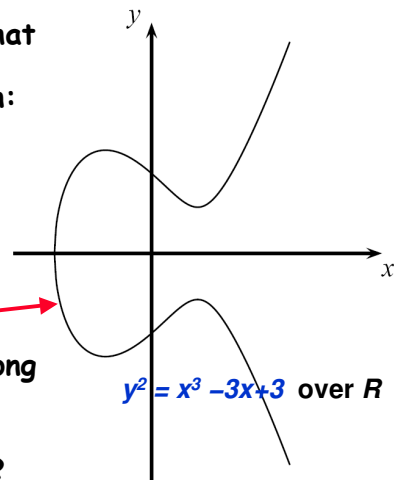
$$y^2 = x^3 + ax + b$$

for parameters a, b that specify the exact shape of the curve

- ◆ On the real numbers and with parameters $a, b \in \mathbb{R}$, an elliptic curve looks like this

- ◆ Elliptic Curves are symmetric along the x -axis

- ◆ Elliptic curves can not just be defined over the real numbers $\mathbb{R}$ but over many other types of finite fields.



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Elliptic Curves over the field Z_p

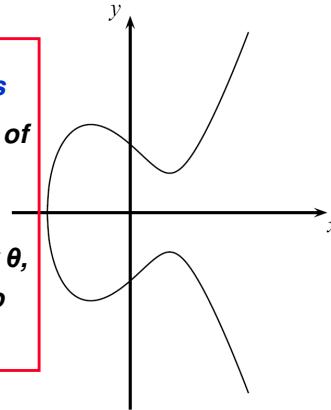
- In cryptography, we are interested in elliptic curves module a prime p :

Definition: Elliptic Curves over prime fields

The elliptic curve E over Z_p ($p > 3$) is the set of all pairs $(x, y) \in Z_p$ which satisfy

$$y^2 = x^3 + ax + b \mod p$$

together with an imaginary point at infinity θ , where $a, b \in Z_p$ and $4a^3 + 27b^2 \neq 0 \mod p$ (to avoid self-intersections of the curve)



- $Z_p = \{0, 1, \dots, p-1\}$ is a set of integers with modulo p arithmetic

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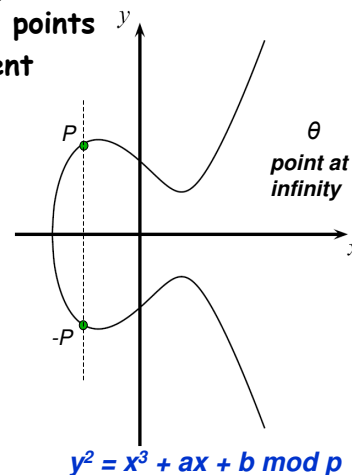
Computations on Elliptic Curves

- Special considerations are required to convert elliptic curves into a **group** of points

- In any group, a special **identity** element is required, i.e.,
given $P \in E$: $P + \theta = P = \theta + P$

- This identity point, denoted by θ is not on the curve and is added to the group definition

- Up to two y and $-y$ exist for each quadratic residue x of the elliptic curve
- For each point $P=(x, y)$, the inverse or negative point is defined as $-P=(x, -y)$
- $P+(-P)=\theta$

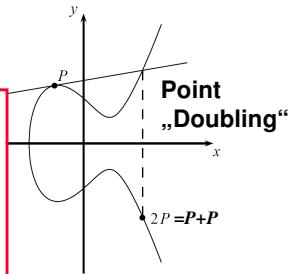
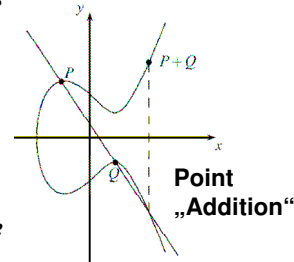


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Computations on Elliptic Curves (ctd.)

- Generating a *group* of points on elliptic curves based on „point addition“ operation $P+Q = W$, i.e., $(x_1, y_1) + (x_2, y_2) = (x_3, y_3)$
- Geometric Interpretation of point addition operation
 - Draw straight line through $P=(x_1, y_1)$ and $Q=(x_2, y_2)$; if $P=Q$ use tangent line instead
 - Mirror 3rd intersection point of drawn line with the elliptic curve with respect to the x -axis
- Elliptic Curve Point Addition and Doubling Formulas



$x_3 = s^2 - x_1 - x_2 \bmod p$ and $y_3 = s(x_1 - x_3) - y_1 \bmod p$
where

$$s = \begin{cases} \frac{y_2 - y_1}{x_2 - x_1} \bmod p & ; \text{ if } P \neq Q \text{ (point addition)} \\ \frac{3x_1^2 + a}{2y_1} \bmod p & ; \text{ if } P = Q \text{ (point doubling)} \end{cases}$$

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Abelian (commutative) group

- $P + \theta = P$
- If $P = (x_1, y_1)$, $-P = (x_1, -y_1)$
- $P + (-P) = \theta$ (not on the curve)
- $P + Q = Q + P$
- $P + (Q + W) = (P + Q) + W$

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Computations on Elliptic Curves - Example 1

- Example: Given $E: y^2 = x^3 + 2x + 2 \pmod{17}$ and point $P = (5, 1)$
Goal: Compute $2P = P + P = (5, 1) + (5, 1) = (x_3, y_3)$

$$s = \frac{3x_1^2 + a}{2y_1} \pmod{p} = (2 \cdot 1)^{-1} (3 \cdot 5^2 + 2) = 2^{-1} \cdot 9 \equiv 9 \cdot 9 \equiv 13 \pmod{17}$$

$$\begin{aligned} x_3 &= s^2 - x_1 - x_2 = 13^2 - 5 - 5 = 159 \equiv 6 \pmod{17} \\ y_3 &= s(x_1 - x_3) - y_1 = 13(5 - 6) - 1 = -14 \equiv 3 \pmod{17} \end{aligned}$$

Finally $2P = (5, 1) + (5, 1) = (6, 3)$

Verify that $(6, 3)$ is a point on the curve:

$$3^2 = 9, 6^3 + 12 + 2 = 36 \cdot 6 + 14 \equiv 2 \cdot 6 + 14 = 26 \equiv 9 \pmod{17}$$

$P + 2P = (5, 1) + (6, 3) = (10, 6)$ since

$$s = \frac{y_2 - y_1}{x_2 - x_1} \pmod{p} = (3 - 1)/(6 - 5) = 2 \cdot (1)^{-1} \equiv 2 \pmod{17}$$

$$x_3 = s^2 - x_1 - x_2 = 4 - 11 = -7 \equiv 10 \pmod{17};$$

$$y_3 = s(x_1 - x_3) - y_1 = -10 - 1 = -11 \equiv 6 \pmod{17}$$

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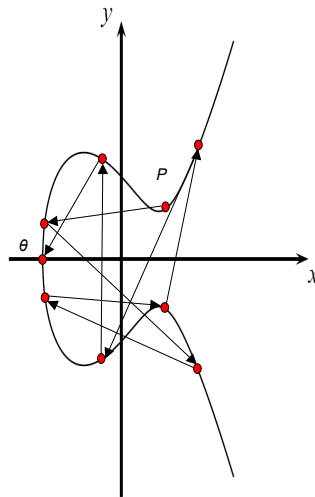
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Example 1 (ctd.)

- The points on an elliptic curve and the point θ form **cyclic** subgroups

$$\begin{array}{ll} 2P = (5, 1) + (5, 1) = (6, 3); & 11P = (13, 10) \\ 3P = 2P + P = (10, 6) & 12P = (0, 11) \\ 4P = (3, 1) & 13P = (16, 4) \\ 5P = (9, 16) & 14P = (9, 1) \\ 6P = (16, 13) & 15P = (3, 16) \\ 7P = (0, 6) & 16P = (10, 11) \\ 8P = (13, 7) & 17P = (6, 14) \\ 9P = (7, 6) & 18P = (5, 16) \\ 10P = (7, 11) & 19P = \theta \end{array}$$

This elliptic curve has order
 $\#E = |E| = 19$ since it contains
 19 points in its cyclic group.



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Example 2

- Given $E: y^2 = x^3 + x + 3 \pmod{7}$
- There are 6 points on this curve: Θ , (4,1), (6,6), (5,0), (6,1), (4,6)
- Addition table:

+	Θ	(4, 1)	(6, 6)	(5, 0)	(6, 1)	(4, 6)
Θ	Θ	(4, 1)	(6, 6)	(5, 0)	(6, 1)	(4, 6)
(4, 1)	(4, 1)	(6, 6)	(5, 0)	(6, 1)	(4, 6)	Θ
(6, 6)	(6, 6)	(5, 0)	(6, 1)	(4, 6)	Θ	(4, 1)
(5, 0)	(5, 0)	(6, 1)	(4, 6)	Θ	(4, 1)	(6, 6)
(6, 1)	(6, 1)	(4, 6)	Θ	(4, 1)	(6, 6)	(5, 0)
(4, 6)	(4, 6)	Θ	(4, 1)	(6, 6)	(5, 0)	(6, 1)

- $P=(4,1)$: $2P=(6,6), 3P=(5,0), 4P=(6,1), 5P=(4,6); 6P=\Theta$
- $P=(4,6)$: $2P=(6,1), 3P=(5,0), 4P=(6,6), 5P=(4,1); 6P=\Theta$
- $P=(5,0)$: $2P=\Theta$; $P=(6,1)$: $2P=(6,6), 3P=\Theta$

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Number of Points on an Elliptic Curve

- ♦ How many points can be on an arbitrary elliptic curve?
 - Example 1, $E: y^2 = x^3 + 2x + 2 \pmod{17}$ has 19 points
 - However, determining the point count on elliptic curves in general is hard
- ♦ But Hasse's theorem bounds the number of points to a restricted interval

Hasse's Theorem:

Given an elliptic curve module p , the number of points on the curve is denoted by $\#E$ and is bounded by

$$p+1-2\sqrt{p} \leq \#E \leq p+1+2\sqrt{p}$$

- ♦ Interpretation: The number of points is „close to“ the prime p
- ♦ Example: To generate a curve with about 2^{160} points, a prime with a length of about 160 bits is required

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Elliptic Curve Discrete Logarithm Problem

- Cryptosystems rely on the hardness of the Elliptic Curve Discrete Logarithm Problem (ECDLP)

Definition: Elliptic Curve Discrete Logarithm Problem (ECDLP)

Given a primitive element P and another element T on an elliptic curve E .

The ECDL problem is finding the integer d , where $1 \leq d \leq \#E$ such that

$$\underbrace{P + P + \dots + P}_{d \text{ times}} = d \times P = T$$

- Cryptosystems are based on the idea that d is large and kept secret and attackers cannot compute it easily
- If d is known, an efficient method to compute the point multiplication $d \times P$ is required to create a reasonable cryptosystem
 - Known Square-and-Multiply Method can be adapted to Elliptic Curves
 - The method for efficient point multiplication on elliptic curves: Double-and-Add Algorithm

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Double-and-Add Algorithm for Point Multiplication

- Double-and-Add Algorithm

Input: Elliptic curve E , an elliptic curve point P and a scalar d with bits d_i

Output: $T = d \times P$

Initialization:

$T = P$

Algorithm:

FOR $i = t - 1$ DOWNTOW 0

$T = T + T \bmod n$

IF $d_i = 1$

$T = T + P \bmod n$

RETURN (T)

Example: $26P = (11010_2)P = (d_4 d_3 d_2 d_1 d_0)_2 P$.

Step

#0	$P = 1_2 P$	initial setting
#1a	$P + P = 2P = 10_2 P$	DOUBLE (bit d_3)
#1b	$2P + P = 3P = 10^2 P + 1_2 P = 11_2 P$	ADD (bit $d_3=1$)
#2a	$3P + 3P = 6P = 2(11_2 P) = 110_2 P$	DOUBLE (bit d_2)
#2b		no ADD ($d_2 = 0$)
#3a	$6P + 6P = 12P = 2(110_2 P) = 1100_2 P$	DOUBLE (bit d_1)
#3b	$12P + P = 13P = 1100_2 P + 1_2 P = 1101_2 P$	ADD (bit $d_1=1$)
#4a	$13P + 13P = 26P = 2(1101_2 P) = 11010_2 P$	DOUBLE (bit d_0)
#4b		no ADD ($d_0 = 0$)

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Elliptic Curve Diffie-Hellman Key Exchange (ECDH)

- Given a prime p , an elliptic curve E and a primitive point $P=(x_p, y_p)$
- The Elliptic Curve Diffie-Hellman Key Exchange protocol:

Alice

Choose $k_{PrA} = a \in \{2, 3, \dots, \#E-1\}$
 Compute $k_{PubA} = A = aP = (x_A, y_A)$

Compute $a \times B = T_{ab}$

Bob

Choose $k_{PrB} = b \in \{2, 3, \dots, \#E-1\}$
 Compute $k_{PubB} = B = bP = (x_B, y_B)$

Compute $b \times A = T_{ab}$

- Joint secret between Alice and Bob: $T_{AB} = (x_{AB}, y_{AB})$
- Proof for correctness:
 - Alice computes $aB = a(bP) = abP$
 - Bob computes $bA = b(aP) = abP$ since group is associative
- One of the coordinates of the point T_{AB} (usually the x -coordinate) can be used as session key

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ECDH (ctd.)

- The ECDH is often used to derive session keys for (symmetric) encryption
- One of the coordinates of the point T_{AB} (usually the x -coordinate) is taken as session key

Alice

Choose $k_{PrA} = a \in \{2, 3, \dots, \#E-1\}$
 Compute $k_{PubA} = A = aP = (x_A, y_A)$

Compute $aB = T_{ab} = (x_T, y_T)$

Define key $k_{AES} = x_T$

Given a message m :
 Encrypt $c = AES_{k_{AES}}(m)$

Bob

Choose $k_{PrB} = b \in \{2, 3, \dots, \#E-1\}$
 Compute $k_{PubB} = B = bP = (x_B, y_B)$

Compute $bA = T_{ab} = (x_T, y_T)$

Define key $k_{AES} = x_T$

Received ciphertext c :
 Decrypt $m = AES_{k_{AES}}^{-1}(c)$

ECDH
Symmetric encryption/decryption

- In some cases, a hash function is used to derive the session key

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Security Aspects

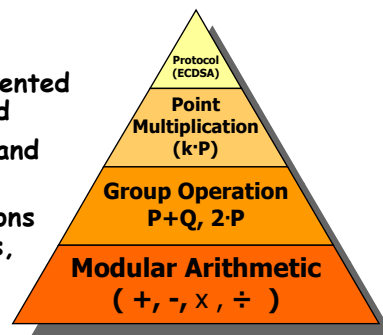
- Why are parameters significantly smaller for elliptic curves (160-256 bit) than for RSA (1024-3076 bit)?
 - Attacks on groups of elliptic curves are weaker than available factoring algorithms or integer DL attacks
 - Best known attacks on elliptic curves (chosen according to cryptographic criterions) are the Baby-Step Giant-Step and Pollard-Rho method
 - Complexity of these methods: on average, roughly $\sqrt{p}$ steps are required before the ECDLP can be successfully solved
- Implications to practical parameter sizes for elliptic curves:
 - An elliptic curve using a prime p with 160 bit (and roughly 2^{160} points) provides a security of 2^{80} steps that required by an attacker (on average)
 - An elliptic curve using a prime p with 256 bit (roughly 2^{256} points) provides a security of 2^{128} steps on average

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Implementations in Hardware and Software

- Elliptic curve computations usually regarded as consisting of four layers:
 - Basic modular arithmetic operations
 - Group operation implements point doubling and point addition
 - Point multiplication can be implemented using the Double-and-Add method
 - Upper layer protocols like ECDH and ECDSA
- Most efforts should go in optimizations of the modular arithmetic operations, such as
 - Modular addition and subtraction
 - Modular multiplication
 - Modular inversion

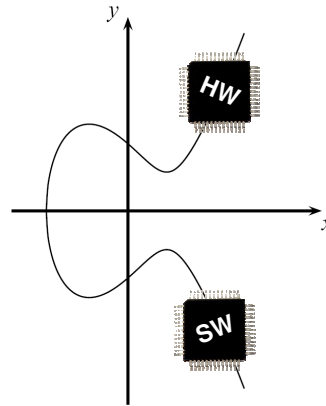


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Implementations in Hardware and Software

- Software implementations
 - Optimized 256-bit ECC implementation on 3GHz 64-bit CPU requires about *2 ms* per point multiplication
 - Less powerful microprocessors (e.g., on SmartCards or cell phones) even take significantly longer (*> 10 ms*)
- Hardware implementations
 - High-performance implementations with 256-bit special primes can compute a point multiplication in a few hundreds microseconds on reconfigurable hardware
 - Dedicated chips for ECC can compute a point multiplication even in a few tens microseconds



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Lessons Learned

- Elliptic Curve Cryptography (ECC) is based on the discrete logarithm problem
- ECC can be used for key exchange, for digital signatures and for encryption
- ECC provides the same level of security as RSA or discrete logarithm systems over $\mathbb{Z}_p$ with considerably shorter operands (approximately 160-256 bit vs. 1024-3072 bit), resulting in shorter ciphertexts and signatures
- In many cases ECC has performance advantages over other public-key algorithms
- ECC is slowly gaining popularity in applications, compared to other public-key schemes, i.e., many new applications, especially on embedded platforms, make use of elliptic curve cryptography

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